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Chapter 1. Introduction

1.1 Information

Information

Information refers to data or knowledge that is meaningful and has value in making decisions or solving problems.

1.1.1 Importance of Information Theory

1.1.1.1 Why Information Theory Matters

- It helps us **quantify** information.
- It helps us **transmit** information efficiently.
- It helps us **optimize the delivery** of information in communication, computing, and biological systems.

1.1.1.2 Real-Life Relevance

- Modern systems such as the **internet**, **mobile networks**, and **social media** rely on efficient data compression and error correction.
- These ideas ensure messages are transmitted **accurately** and **efficiently**.
- Information theory also helps reduce **noise interference** in signals.
- This improves reliability in applications such as **streaming**, **medical diagnostics**, and **genomic analysis**.

1.1.2 Importance in Communication

1.1.2.1 Key Points

- Information theory provides mathematical frameworks to optimize **data transmission** and **storage**.
- It helps determine the maximum amount of information a channel can handle, that is, **channel capacity**.
- It helps determine how to encode messages so that transmission errors can be reduced.

1.1.2.2 Communication Insight

- Cellular networks and internet protocols use **error correction** and **data compression**.
- These make communication **efficient** and **reliable**.
- Even in noisy environments, users can still experience:
  - clear calls,
  - fast internet speeds,

- accurate data exchange.

This is why information theory is fundamental to seamless global communication.

1.2 Discrete Source

Discrete Source

A discrete source is a system that generates a **finite set of symbols or values** from a **predefined set of alphabets**. These symbols are discrete and distinct, such as letters, numbers, or pixels.

1.2.0.1 Examples of Discrete Sources

- **Text:** A keyboard generates letters, numbers and special characters.
- **Audio:** Digital audio files sample sound at specific intervals, producing discrete data points.
- **Image/Video:** Images are composed of pixels with distinct color values, while videos are sequences of discrete image frames.

1.2.1 Source Types

1.2.1.1 Memoryless Source

Memoryless Source

A discrete source is memoryless if the symbols it generates are **independent** of each other.

- The current symbol has **no dependence** on past symbols.
- Example: flipping a fair coin repeatedly produces independent outcomes (H or T).
- Such sources simplify analysis because each symbol can be treated individually without considering historical context.
- Example:A random number generator producing digits (0–9) where each number is equally likely and independent of the previous numbers.

1.2.1.2 Memory Source

Memory Source

A discrete source is a memory source if the generation of a symbol **depends** on one or more **previous symbols**.

- Such sources have **statistical dependency** between symbols.
- This dependency can be modeled using **Markov processes** or **conditional probabilities**.
- They are common in **natural language and encoded data streams**, where rules and structure influence symbol generation.
- Example: In English text, the letter ‘q’ is almost always followed by ‘u’, showing a strong dependency between these symbols.

1.3 Self Information

Self Information

Self information measures the **amount of uncertainty or surprise** associated with the occurrence of a specific event. It quantifies how much **information is gained** when an event occurs.

Self Information Formula

Consider a discrete random variable X with possible outcomes x_i, $i = 1, 2, 3, \ldots, n$. Self information of the event $X = x_i$ is defined as

$$I(x_i) = \log \frac{1}{P(x_i)} = -\log P(x_i).$$

- $I(x_i)$ = self information of the event x_i
- $P(x_i)$ = probability of the event x_i

1.3.1 Properties of Self Information

1. If we are absolutely certain of the outcome of an event even before it occurs, there is no information gained.

$$I(x_i) = 0, \quad \text{if } P_i = 1$$

This means **certainty carries no message information**.

2. The occurrence of an event either provides some or no information, but never brings about a loss of information.

$$I(x_i) \geq 0, \quad \text{for } 0 \leq P(x_i) \leq 1$$

That implies **self information cannot be negative**.

3. The **less probable an event is, the more information we gain** when it occurs.

$$I(x_1) > I(x_2) \quad \text{if } P_1 < P_2$$

4. For statistically independent events, the total information obtained from the joint occurrence is equal to the sum of the individual informations.

$$I(x_1, x_2) = I(x_1) + I(x_2).$$

This means that **information is additive for independent events**.

Equivalently,

$$I_A + I_B = \log \frac{1}{P_A} + \log \frac{1}{P_B} = \log \frac{1}{P_A P_B} = \log \frac{1}{P(AB)}.$$

This follows because for independent events,

$$P(AB) = P_A \cdot P_B.$$

Thus, the information of the joint event can be written as

$$I(AB) = \log \frac{1}{P(AB)},$$

which matches the sum of the individual informations. This implies that **independent events do not share information, so their informations simply add up**.

1.3.2 Physical significance / insight of self information

Consider a set of events $X = \{x_1, x_2, x_3\}$ with probability

$$P = \{0.5, 0.25, 0.25\}.$$

Step 1: Self information for the first event:

$$I_{x_1} = \log_2 \frac{1}{0.5} = -\log_2 0.5 = 1 \text{ bit}$$

Step 2: For the other two events:

$$I_{x_2} = I_{x_3} = \log_2 \frac{1}{0.25} = -\log_2 0.25 = 2 \text{ bit}$$

Observation

- Higher probability requires fewer bits.
- Lower probability requires more bits.
- Lower probability therefore means a higher degree of uncertainty.

1.4 Entropy

Entropy

In information theory, entropy is the measurement of the average amount of information, that is, the number of bits of uncertainty, in a set of possible outcomes per symbol.

1.4.1 Interpretation

- It measures the **unpredictability** of a random variable.
- It represents the minimum number of bits required to encode the information generated by a source.

Entropy of an Event

$$H(x_i) = -P_i \log_2 P_i.$$

- $H(x_i)$ = entropy contribution of the event x_i
- P_i = probability of the ith event

1.4.2 Why This Form Appears

- Self information of an event is $(-\log_2 P_i)$.
- Entropy is an **average** information measure.
- So self information is weighted by its probability (P_i).

Entropy of a Random Variable

$$H(X) = -\sum_{i=1}^{n} P_i \log_2 P_i \quad \text{bit/symbol.}$$

- $H(X)$ = average entropy of the random variable X
- n = total number of possible outcomes
- P_i = probability of the ith outcome

Example 1.1

A source generates information with probabilities

$$P_1 = 0.1, \quad P_2 = 0.2, \quad P_3 = 0.3, \quad P_4 = 0.4.$$

Find the entropy of the system and what percentage of maximum possible information is being generated by this source.

Step 1: Calculate entropy

$$H(X) = -\sum_{i=1}^{n} P_i \log_2 P_i$$

$$= -\Big(0.1\log_2(0.1) + 0.2\log_2(0.2) + 0.3\log_2(0.3) + 0.4\log_2(0.4)\Big)$$

$$= 1.8464 \approx 2 \text{ bits/symbol.}$$

Step 2: Calculate maximum entropy

$$H_{\max} = \log_2 4 = 2.$$

Step 3: Calculate efficiency

$$\eta = \frac{H}{H_{\max}} \times 100 = \frac{1.8464}{2} \times 100 = 92.32\%.$$

Question 1.2

Q: Why is P_i multiplied with self information in the entropy equation?
A:

- Entropy is an *average* measure of information.
- So the self information term $(-\log_2 P_i)$ must be weighted by its probability P_i.
- This keeps highly probable events contributing more frequently and less probable events contributing less frequently to the final average.
- It also keeps the entropy value bounded properly for a valid probability distribution. For example- the probability must be in between 0 and 1. If -

$$\begin{aligned} P = 0.1 &\Rightarrow I = -\log_2(0.1) = 3.32 \text{ bits (more than 1 bit)} \\ P = 0.5 &\Rightarrow I = -\log_2(0.6) = 0.74 \text{ bit} \\ P = 0.9 &\Rightarrow I = -\log_2(0.9) = 0.152 \text{ bits} \end{aligned}$$

  To maintain entropy value below 1 bit/symbol, we need to multiply the self information with the probability P_i.

1.4.3 Significance of Entropy

- **Measures uncertainty:** quantifies unpredictability in a source, providing insight into the information content.
- **Data compression:** sets the theoretical limits for lossless compression, helping design efficient encoding systems.
- **Error detection/correction:** guides the design of coding techniques to handle noise in communication channels.
- **System efficiency:** helps evaluate and optimize the performance of communication and storage systems.
- **Applications in AI/ML:** used in decision trees and deep learning for feature selection and model optimization.

1.4.4 Relation between Hartley, Nats and Bits

1.4.4.1 Units of Self Information

- Self information in Hartley:

$$I = \log_{10} \frac{1}{P}$$

- Self information in nats:

$$I = \log_e \frac{1}{P}$$

- Self information in bits:

$$I = \log_2 \frac{1}{P}$$

Hartley to Bits

$$1 \text{ Hartley} = \frac{I}{-\log_{10} P} = \frac{-\log_2 P}{-\log_{10} P} = \frac{\log 10}{\log 2} = \log_2 10 = 3.3219 \text{ bits.}$$

Hartley to Nats

$$1 \text{ Hartley} = \frac{I}{-\log_{10} P} = \frac{-\log_e P}{-\log_{10} P} = \frac{\log 10}{\log e} = \log_e 10 = 2.302 \text{ nats.}$$

Nats to Bits

$$1 \text{ Nat} = \frac{I}{-\log_e P} = \frac{-\log_2 P}{-\log_e P} = \frac{\log e}{\log 2} = \log_2 e = \frac{1}{\ln 2} = 1.442 \text{ bits.}$$

Example 1.3

Q: If there are 100 equiprobable outcomes, calculate the corresponding values in nats, bits and Hartley.
A: Given $N = 100$, for equiprobable outcomes:

$$P(x_i) = \frac{1}{100}$$

Using entropy formula:

$$H(X) = -\sum_{i=1}^{100} \frac{1}{100} \log \frac{1}{100}$$

Since all terms are identical:

$$H(X) = -100 \cdot \frac{1}{100} \log \frac{1}{100}$$

$$H(X) = -\log \frac{1}{100}$$

Now simplify:

$$H(X) = \log 100$$

Concept:

- Bits, nats, and Hartley are **units of information** (log base), used for both $I(x)$ and $H(X)$.
- The question refers to a **set of outcomes**, so we use **entropy**:

$$H(X) = -\sum P(x_i) \log P(x_i)$$

- For equiprobable case:

$$I(x_i) = -\log \frac{1}{100} = \log 100, \quad H(X) = \log 100$$

- Hence, **self-information = entropy** in this case.
- Units depend on log base:

$$\log_2 \to \text{bits}, \quad \log_e \to \text{nats}, \quad \log_{10} \to \text{Hartley}$$

Note: Bits, nats and Hartley are units of information (log base).

$$\text{In bits, } H(X) = \log_2(100) = 6.64$$
$$\text{In nats, } H(X) = \log_e(100) = 4.605$$
$$\text{In Hartley, } H(X) = \log_{10}(100) = 2$$

1.4.5 Equal Probability Leads to High Entropy

Consider a BPSK output

$$x = \begin{cases} 1, & \text{probability } p \\ 0, & \text{probability } 1-p \end{cases}$$

$$H(x) = -\sum_{x=0}^{1} P(x) \log_2 P(x)$$

$$= -p \log_2 p - (1-p) \log_2 (1-p).$$

- p = probability of symbol 1
- $1-p$ = probability of symbol 0
- $H(x)$ = entropy of the binary source

  For equal probability $p = 0.5$ and $1 - p = 0.5$,

$$H(x) = -0.5 \log_2 0.5 - 0.5 \log_2 0.5 = 1 \text{ bit/symbol.}$$

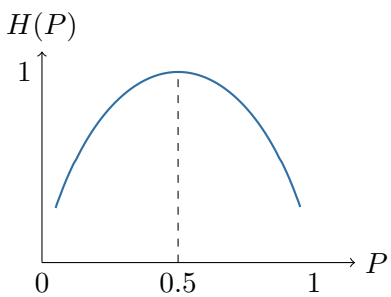

Figure 1.1: Entropy of a binary source versus probability

From the figure, it is clear that for $p = 0$ or $p = 1$, entropy $H(x)$ is zero, but for $p = 0.5$, entropy is maximum. Equal probability leads to the highest uncertainty, so it also leads to the highest entropy.

Example 1.4

Calculate the entropy of the QPSK transmitted signal below:

00 01 10 10 11 11 01 01 10 00 10 10 11 01 10

QPSK has four distinct $(00, 01, 10, 11)$ bit combinations. Thus, $2^n = 4 \Rightarrow n = 2$. So, we have to take two bits for each symbol.

Symbol	Times
00	2
01	4
10	6
11	3

$$P_{00} = \frac{2}{15}, \qquad P_{01} = \frac{4}{15}, \qquad P_{10} = \frac{6}{15}, \qquad P_{11} = \frac{3}{15}$$

$$\begin{aligned} H(X) &= -\left[P_{00}\log_2 P_{00} + P_{01}\log_2 P_{01} + P_{10}\log_2 P_{10} + P_{11}\log_2 P_{11}\right] \\ &= -\left[\frac{2}{15}\log_2\frac{2}{15} + \frac{4}{15}\log_2\frac{4}{15} + \frac{6}{15}\log_2\frac{6}{15} + \frac{3}{15}\log_2\frac{3}{15}\right] \\ &= 1.889 \approx 2 \text{ bits/symbol.} \end{aligned}$$

Physical significance: Thus, we need at least 2 bits to transmit the signal.

1.5 Joint Entropy

Joint Entropy

The joint entropy $H(X,Y)$ of a pair of random variables (X,Y) with joint distribution $P(x,y)$ is the average uncertainty associated with the pair of outcomes taken together.

Joint Entropy Formula

$$H(X,Y) = -\sum_{x\in X}\sum_{y\in Y} P(x,y)\log_2 P(x,y)$$
$$= -E\log_2 P(x,y)$$

- $H(X,Y)$ = joint entropy of the variables X and Y
- $P(x,y)$ = joint probability of the pair (x,y)
- $E[\cdot]$ = expectation

Example 1.5

Two variables are considered for a student:

- X = study hours
- Y = attendance

Their joint probabilities are listed below. Show that the variables are dependent on each other.

X	**Y**	**P(X, Y)**
Low	Low	0.20
Low	High	0.10
High	Low	0.15
High	High	0.55

Step 1: Joint entropy contribution of each case

$$H_{LL}(X,Y) = -0.2\log_2(0.2) = 0.46$$
$$H_{LH}(X,Y) = -0.1\log_2(0.1) = 0.33$$
$$H_{HL}(X,Y) = -0.15\log_2(0.15) = 0.41$$
$$H_{HH}(X,Y) = -0.55\log_2(0.55) = 0.4743$$

$$H(X,Y) = 0.46 + 0.33 + 0.41 + 0.4743 = 1.6743$$

Step 2: Marginal entropy for study hours

$$P_L = 0.2 + 0.1 = 0.3, \qquad P_H = 0.15 + 0.55 = 0.7$$

$$H(X) = -\left[0.3\log_2(0.3) + 0.7\log_2(0.7)\right] = 0.881$$

Step 3: Marginal entropy for attendance

$$P_L = 0.2 + 0.15 = 0.35, \qquad P_H = 0.1 + 0.55 = 0.65$$

$$H(Y) = -\left[0.35\log_2(0.35) + 0.65\log_2(0.65)\right] = 0.934$$

Conclusion

$$H(X) + H(Y) = 0.881 + 0.934 = 1.815$$

Since $H(X) + H(Y) > H(X,Y)$, the variables X and Y are dependent. We can also say that the variables contain mutual information.

1.6 Mutual Information

Mutual Information

Mutual information measures the amount of information that one random variable X provides about another random variable Y. It quantifies the reduction in uncertainty about one variable when the other is known.

Mutual Information Formula

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

$$\text{or equivalently,} \qquad I(X;Y) = H(X) - H(X \mid Y)$$

- $I(X;Y)$ = mutual information between X and Y
- $H(X)$, $H(Y)$ = marginal entropies
- $H(X,Y)$ = joint entropy
- $H(X \mid Y)$ = conditional entropy of X given Y

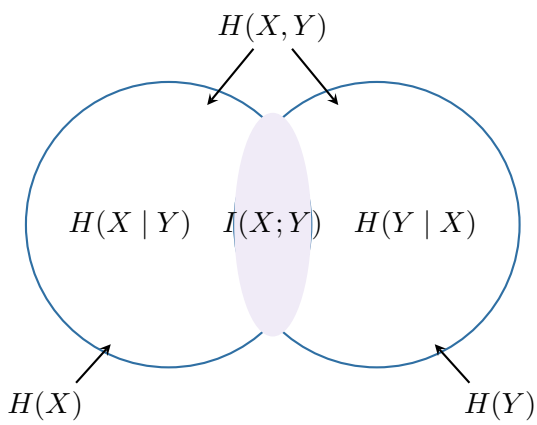

Figure 1.2: Mutual information as the overlap between $H(X)$ and $H(Y)$

Example 1.6

Calculate the joint entropy for the following data and also find the mutual information.

Y \ X	0	1
0	$\frac{1}{2}$	$\frac{1}{4}$
1	$\frac{1}{8}$	$\frac{1}{8}$

Step 1: Joint entropy

$$\begin{aligned} H(X,Y) &= -\left[\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4} + \frac{1}{8}\log_2\frac{1}{8} + \frac{1}{8}\log_2\frac{1}{8}\right] \\ &= 1.75 \end{aligned}$$

Step 2: Entropy of Y

$$P(Y=0) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}, \qquad P(Y=1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

$$H(Y) = \frac{3}{4}\log_2\frac{4}{3} + \frac{1}{4}\log_2 4 = 0.811$$

Step 3: Entropy of X

$$P(X=0) = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}, \qquad P(X=1) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$$

$$H(X) = \frac{5}{8}\log_2\frac{8}{5} + \frac{3}{8}\log_2\frac{8}{3} = 0.954$$

Step 4: Conditional entropies

$$H(X \mid Y) = H(X,Y) - H(Y) = 1.75 - 0.811 = 0.939$$

$$H(Y \mid X) = H(X,Y) - H(X) = 1.75 - 0.954 = 0.796$$

Step 5: Mutual information

$$I(X;Y) = H(X) + H(Y) - H(X,Y) = 0.954 + 0.811 - 1.75 = 0.015$$

1.7 Conditional Entropy

Conditional Entropy

Conditional entropy measures the amount of uncertainty remaining in a random variable when another related variable is already known. It tells us how much additional information is still needed to describe X when Y is already provided.

1.7.1 Simple Intuition

Consider the set $\{1, 2, 3, 4, 5, 6\}$. If we only know whether the outcome is odd or even, some uncertainty still remains.

- If the outcome is **odd**, the possible values are $\{1, 3, 5\}$.
- If the outcome is **even**, the possible values are $\{2, 4, 6\}$.

This remaining uncertainty is the idea captured by conditional entropy.

Conditional Entropy Formula

$$\begin{aligned} H(Y \mid X) &= -\sum_{x \in X} P(x) \sum_{y \in Y} P(y \mid x) \log_2 P(y \mid x) \\ &= -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(y \mid x) \end{aligned}$$

- $H(Y \mid X)$ = uncertainty of Y after X is known
- $P(y \mid x)$ = conditional probability of $Y = y$ given $X = x$
- $P(x, y)$ = joint probability of X and Y

Useful Relation for the Derivation

$$P(y \mid x) = \frac{P(x, y)}{P(x)}$$

$$\log(AB) = \log A + \log B$$

- The first relation rewrites conditional probability in terms of joint and marginal probability.
- The logarithm identity is used when expanding $\log_2\big(P(x)P(y \mid x)\big)$.

1.7.2 Relation with Joint Entropy

Starting from the joint entropy,

$$\begin{aligned} H(X, Y) &= -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(x, y) \\ &= -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 \big(P(x) P(y \mid x)\big) \\ &= -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(x) - \sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(y \mid x) \\ &= -\sum_{x \in X} P(x) \log_2 P(x) - \sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(y \mid x) \end{aligned}$$

Hence,

$$H(X, Y) = H(X) + H(Y \mid X)$$

Similarly,

$$H(X, Y) = H(Y) + H(X \mid Y)$$

1.8 Relative Entropy

Relative Entropy

Relative entropy, also called Kullback-Leibler divergence, measures how different an estimated probability distribution is from the true distribution.

What Relative Entropy Means

- Relative entropy compares two probability mass functions: $P(x)$ and $Q(x)$.
- $P(x)$ represents the true or actual distribution.
- $Q(x)$ represents the estimated or assumed distribution.
- It tells us how far the estimate $Q(x)$ is from the true distribution $P(x)$.
- A larger value of $D(P \parallel Q)$ means the deviation from the actual distribution is higher.
- If the two distributions are identical, then the relative entropy becomes zero.

Kullback-Leibler Divergence

$$D(P \parallel Q) = \sum_{x \in X} P(x) \log_2 \frac{P(x)}{Q(x)}$$

- $P(x)$ = true probability mass function
- $Q(x)$ = estimated probability mass function
- $D(P \parallel Q)$ = relative entropy from Q to P

Basic Points

- If $P(x) = Q(x)$, then $D(P \parallel Q) = 0$.

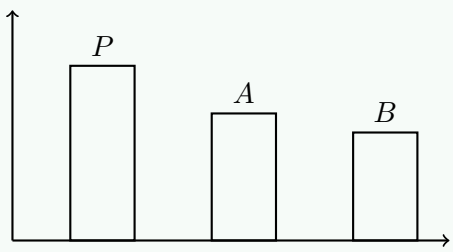

Figure 1.3: Relative entropy illustration comparing the true distribution P with estimates A and B

- In the illustration:

$$D(P \parallel P) = 0, \qquad D(P \parallel A) = 0.25, \qquad D(P \parallel B) = 1.25$$

- Higher relative entropy means the estimated distribution deviates more from the actual distribution.

Example 1.7

Consider a scenario where the true distribution is

$$P_{\text{dog}} = 1, \qquad P_{\text{cat}} = 0$$

For model A,

$$P'_{\text{dog}} = 0.1, \qquad P'_{\text{cat}} = 0.9$$

For model B,

$$P'_{\text{dog}} = 0.9, \qquad P'_{\text{cat}} = 0.1$$

Find the relative entropy using KL divergence.

For model A:

$$D(P \parallel P'_A) = P_{\text{dog}} \log_2 \frac{P_{\text{dog}}}{P'_{\text{dog}}} + P_{\text{cat}} \log_2 \frac{P_{\text{cat}}}{P'_{\text{cat}}}$$

$$= 1 \log_2 \frac{1}{0.1} + 0 \cdot \log_2 \frac{0}{0.9} = \log_2 10 = 3.32 \text{ unit}$$

For model B:

$$D(P \parallel P'_B) = P_{\text{dog}} \log_2 \frac{P_{\text{dog}}}{P'_{\text{dog}}} + P_{\text{cat}} \log_2 \frac{P_{\text{cat}}}{P'_{\text{cat}}}$$

$$= 1 \log_2 \frac{1}{0.9} + 0 \cdot \log_2 \frac{0}{0.1} = 0.15 \text{ unit}$$

So model B is better because it has lower KL divergence.

1.8.1 Mutual Information as Relative Entropy

For two dependent random variables (X, Y), mutual information can be written as the relative entropy between the joint distribution $P(x, y)$ and the product of the marginals $P(x)P(y)$:

Mutual Information from Relative Entropy

$$I(X;Y) = \sum_{x,y} P(x,y) \log_2 \frac{P(x,y)}{P(x)P(y)} = D\big(P(x,y) \,\|\, P(x)P(y)\big)$$

- $I(X;Y)$ = mutual information between X and Y
- $P(x,y)$ = joint probability distribution of X and Y
- $P(x)$, $P(y)$ = marginal probability distributions of X and Y
- $P(x)P(y)$ = product of marginals, representing the independent case
- $D\big(P(x,y) \,\|\, P(x)P(y)\big)$ = relative entropy between the joint distribution and the product of marginals

$$\begin{aligned}
I(X;Y) &= \sum_{x,y} P(x,y) \log_2 \frac{P(x,y)}{P(x)P(y)} \\
&= \sum_{x,y} P(x,y) \log_2 \frac{P(x)P(y \mid x)}{P(x)P(y)} \\
&= \sum_{x,y} P(x,y) \log_2 \frac{P(y \mid x)}{P(y)} \\
&= \sum_{x,y} P(x,y) \log_2 P(y \mid x) - \sum_{x,y} P(x,y) \log_2 P(y) \\
&= -H(Y \mid X) + H(Y) \\
\text{So, } I(X;Y) &= H(Y) - H(Y \mid X)
\end{aligned}$$

Also,

$$H(X,Y) = H(X) + H(Y \mid X)$$

Therefore,

$$H(Y \mid X) = H(X,Y) - H(X)$$

Substituting this into the previous expression,

$$I(X;Y) = H(Y) - H(X,Y) + H(X)$$

Thus the mutual information formula becomes

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

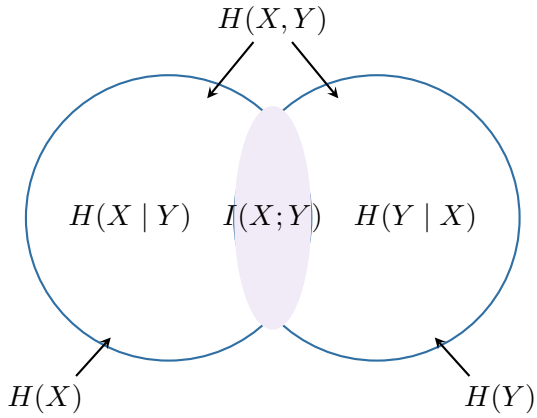

Figure 1.4: Mutual information as the overlap between $H(X)$ and $H(Y)$

1.8.2 Convexity and Jensen's Inequality

Convex Function

A function f is convex over an interval if its second derivative is non-negative over that interval:

$$\frac{d^2 f(x)}{dx^2} \geq 0$$

Examples: x^2 and e^x.

Consider the function $f(x) = x^2$ with $a = 0$ and $b = 2$.

- $f(0) = 0$ and $f(2) = 4$
- Average of the function values:
$$\frac{f(0) + f(2)}{2} = \frac{0 + 4}{2} = 2$$
- Function of the average:
$$f\left(\frac{a+b}{2}\right) = f\left(\frac{0+2}{2}\right) = f(1) = 1^2 = 1$$

Key Observation

For a convex function, the average of the function is greater than or equal to the function of the average:

$$\text{Average of the function} \geq \text{Function of the average}$$

Jensen's Inequality

If a function $f(x)$ is convex over an interval (a, b), then for every $x_1, x_2 \in (a, b)$ and $0 \leq \lambda \leq 1$,

$$\lambda f(x_1) + (1 - \lambda) f(x_2) \geq f\big[\lambda x_1 + (1 - \lambda) x_2\big]$$

A function f is said to be strictly convex if equality holds only when $\lambda = 0$ or $\lambda = 1$.

Example 1.8

For $x_1 = 0$, $x_2 = 2$, $\lambda = 0.5$, and $f(x) = x^2$,

$$\lambda f(x_1) + (1 - \lambda) f(x_2) \geq f\big[\lambda x_1 + (1 - \lambda) x_2\big]$$

$$0.5 \times 0 + 0.5 \times 4 \geq f[0.5 \times 0 + 0.5 \times 2]$$

$$2 \geq 1$$

Thus $f(x) = x^2$ is a convex function.

1.8.3 Gibbs Information Inequality

Statement

Let $P(x)$ and $Q(x)$, where $x \in X$, be two probability mass functions. Then

$$D(P \parallel Q) \geq 0$$

with equality if and only if $P(x) = Q(x)$ for all x.

Let

$$A = \{x : P(x) > 0\}$$

be the support set of $P(x)$. Then

$$\begin{aligned}
-D(P \parallel Q) &= -\sum_{x \in A} P(x) \log \frac{P(x)}{Q(x)} \\
&= \sum_{x \in A} P(x) \log \frac{Q(x)}{P(x)} \\
&\leq \log \sum_{x \in A} P(x) \frac{Q(x)}{P(x)} \\
&= \log \sum_{x \in A} Q(x) \\
&\leq \log \sum_{x \in X} Q(x) \\
&= \log 1 \\
&= 0
\end{aligned}$$

The inequality step follows because $\log(\cdot)$ is a strictly concave function.
Therefore,

$$D(P \parallel Q) \geq 0$$

and relative entropy becomes zero only when the two distributions are identical.

1.8.4 Example: Asymmetric Nature of Relative Entropy

Example 1.9

Let $X = \{0, 1\}$ and consider two distributions P and Q on X:

$$P(0) = 1 - r, \qquad P(1) = r, \qquad Q(0) = 1 - s, \qquad Q(1) = s$$

Find $D(P \parallel Q)$ and $D(Q \parallel P)$.
We have

$$D(P \parallel Q) = \sum_x P(x) \log_2 \frac{P(x)}{Q(x)} = P(0) \log_2 \frac{P(0)}{Q(0)} + P(1) \log_2 \frac{P(1)}{Q(1)}$$
$$= (1 - r) \log_2 \frac{1 - r}{1 - s} + r \log_2 \frac{r}{s}$$

When $r = s = \frac{1}{2}$, then

$$D(P \parallel Q) = 0$$

When $r = \frac{1}{2}$ and $s = \frac{1}{4}$,

$$D(P \parallel Q) = \frac{1}{2} \log_2 \frac{1 - \frac{1}{2}}{1 - \frac{1}{4}} + \frac{1}{2} \log_2 \frac{\frac{1}{2}}{\frac{1}{4}}$$
$$= \frac{1}{2} \log_2 \frac{\frac{1}{2}}{\frac{3}{4}} + \frac{1}{2} \log_2 \frac{\frac{1}{2}}{\frac{1}{4}}$$
$$= -0.292481 + \frac{1}{2} = 0.2075 \text{ bit}$$

Similarly,

$$D(Q \parallel P) = (1 - s) \log_2 \frac{1 - s}{1 - r} + s \log_2 \frac{s}{r}$$
$$= \left(1 - \frac{1}{4}\right) \log_2 \frac{1 - \frac{1}{4}}{1 - \frac{1}{2}} + \frac{1}{4} \log_2 \frac{\frac{1}{4}}{\frac{1}{2}}$$
$$= \frac{3}{4} \log_2 \frac{\frac{3}{4}}{\frac{1}{2}} + \frac{1}{4} \log_2 \frac{1}{2}$$
$$= 0.438721 - 0.25$$
$$= 0.188721 \text{ bit}$$

Hence,

$$D(P \parallel Q) \neq D(Q \parallel P)$$

So the asymmetric relationship is proved.

1.9 Chain Rule

1.9.1 Entropy

Chain Rule for Entropy

$$H(X_1, X_2, \ldots, X_n) = \sum_{i=1}^{n} H(X_i \mid X_1, \ldots, X_{i-1})$$

- For $n = 2$:

$$H(1, 2) = H(1) + H(2 \mid 1)$$

- For $n = 3$:

$$H(1, 2, 3) = H(1) + H(2 \mid 1) + H(3 \mid 1, 2)$$

1.9.2 Conditional Entropy

Chain Rule for Conditional Entropy

$$H(X_1, X_2, \ldots, X_n \mid Y) = \sum_{i=1}^{n} H(X_i \mid X_1, \ldots, X_{i-1}, Y)$$

1.9.3 Mutual Information

Chain Rule for Mutual Information

$$I(X_1, X_2, \ldots, X_n; Y) = \sum_{i=1}^{n} I(X_i; Y \mid X_1, \ldots, X_{i-1})$$

- For $n = 2$:

$$I(1, 2; Y) = I(1; Y) + I(2; Y \mid 1)$$

- For $n = 3$:

$$I(1, 2, 3; Y) = I(1; Y) + I(2; Y \mid 1) + I(3; Y \mid 1, 2)$$

1.9.4 Worked Examples

Entropy of a Finite-Length DMS 1.10

Calculate the entropy of the finite-length DMS X with probability distribution

$$P(x) = \{0.125, 0.25, 0.5, 0.125\}, \qquad x \in X$$

Entropy,

$$H(X) = \sum_{x \in X} H(x_i) = -\sum_{i=1}^{n} p_i \log_2 p_i$$

$$= -\left[2(0.125)\log_2(0.125) + (0.25)\log_2(0.25) + (0.5)\log_2(0.5)\right]$$

$$= 1.375$$

Average Self Information of a Source 1.11

For any source X with pmf

$$P(x) = \{0.0625, 0.5, 0.25, 0.125, 0.0625\}, \qquad x \in X$$

calculate its average self information.
The average self information is nothing but the entropy.

$$\begin{aligned} H(X) &= -\left[2 \times 0.0625 \log_2 0.0625 + 0.5 \log_2 0.5 + 0.25 \log_2 0.25 + 0.125 \log_2 0.125\right] \\ &= 1.875 \end{aligned}$$

1.10 Worked Problems

Fully Dependent System 1.12

Prove that for a fully dependent system, mutual information is the same as self information.
Let two random variables X and Y be fully dependent. Knowing X gives full knowledge of Y, and vice versa. Therefore,

$$H(Y \mid X) = 0$$

From the formula of mutual information,

$$I(X;Y) = H(Y) - H(Y \mid X)$$

$$= H(Y) - 0$$

So,

$$I(X;Y) = H(Y)$$

Hence, mutual information equals the entropy, that is, the average self information. All uncertainty in one variable is resolved by knowing the other.

8-PSK Modulator Example 1.13

An 8-PSK modulator generates the following symbol sequence:

010 110 111 101 011 000 011 101 110

101 110 111 010 001 101 100

Determine the average information of the modulator.
The average information is the entropy.

Symbol	Count
000	1
001	1
010	2
011	2
100	1
101	4
110	3
111	2

$$H = -\left[3 \times \frac{1}{16}\log_2\frac{1}{16} + 3 \times \frac{2}{16}\log_2\frac{2}{16} + \frac{3}{16}\log_2\frac{3}{16} + \frac{4}{16}\log_2\frac{4}{16}\right]$$

$$= \frac{3}{4} + \frac{9}{8} + 0.4528195311 + \frac{1}{2}$$

$$\approx 2.8278 \text{ bits/symbol}$$

Significance: At least 3 bits are required to transmit the 8-PSK modulated signal.

1.11 Channel Interpretation

Mutual Information as a Measure of Channel Goodness 1.14

Mutual information can be interpreted as a measure of channel goodness. Justify the statement with an example.

Answer

- Mutual information $I(X;Y)$ quantifies how much information the received signal Y reveals about the transmitted signal X.
- A higher mutual information means less uncertainty in decoding and a more reliable channel.
- For a perfect channel, $I(X;Y) = H(X)$. All transmitted information is received without loss.
- For a noisy channel, $I(X;Y) = 0$. The output reveals nothing about the input, indicating a useless or fully noisy channel.

Example

For a binary symmetric channel with low crossover probability p, $I(X;Y)$ is high, showing good channel reliability. For BSC, let $p = 0.1$.

$$\begin{aligned} C &= \max_{P(X)} I(X;Y) = 1 - H(p) \\ &= 1 - [-p\log_2 p - (1-p)\log_2(1-p)] \\ &= 1 + 0.1\log_2(0.1) + 0.9\log_2(0.9) \\ &= 0.531 \text{ bit per bit} \end{aligned}$$

The channel is fairly good but not perfect.

Probability of Correct Reception 1.15

Prove that the probability of correct reception is $1 - P_e$.

Answer

Let P_c be the probability of correct reception and P_e be the probability of error.
Since these are complementary events,

$$P_c + P_e = 1$$

or,

$$P_c = 1 - P_e$$

Hence proved.

1.12 Discrete Memoryless Channel

Discrete Memoryless Channel (DMC)

A discrete memoryless channel (DMC) is a channel where:

- the input and output alphabets are discrete (finite),
- the current output depends only on the current input,

- the current output does not depend on any previous inputs or outputs.

So, the channel has no memory.

Channel Matrix

The channel matrix of a DMC is a table showing the conditional probabilities of receiving output symbols given an input symbol.

Example 1.16

Let the input alphabet be $\{0, 1\}$ and the output alphabet be $\{0, 1\}$. If

$$P(y = 0 \mid x = 0) = 0.9, \qquad P(y = 1 \mid x = 0) = 0.1$$

$$P(y = 1 \mid x = 1) = 0.8, \qquad P(y = 0 \mid x = 1) = 0.2$$

then the channel matrix is

$$\begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

1.12.1 Average Information Rate of a Source

Worked Example 1.17

The output of an information source consists of 150 symbols, 32 of which occur with a probability of $\frac{1}{64}$, and the remaining with a probability of $\frac{1}{236}$. The source emits 2000 symbols/sec. Assume the symbols are chosen independently. Find the average information rate of this source.

Given

- $p_1 = \frac{1}{64}$
- $p_2 = \frac{1}{236}$
- $n_1 = 32$
- $n_2 = 150 - 32 = 118$

Total Entropy

$$\begin{aligned} H(X) &= -\left(p_1 \log_2 p_1 \times n_1 + p_2 \log_2 p_2 \times n_2\right) \\ &= -\left(\frac{1}{64} \log_2 \frac{1}{64} \times 32 + \frac{1}{236} \log_2 \frac{1}{236} \times 118\right) \\ &= 6.9413 \text{ bit/symbol} \end{aligned}$$

Average Information Rate

$$\begin{aligned} R &= H(X) \times r \\ &= 6.9413 \times 2000 \\ &= 13882.64305 \text{ bits/sec} \end{aligned}$$

1.13 Additional Worked Problems

Maximum Entropy for Two Messages 1.18

Consider a source that transmits two independent messages with probabilities p and $(1-p)$, respectively. Demonstrate that the entropy is maximized when the two messages are equally likely. Plot the entropy of source vs probability.

Answer

For equally likely messages, the probabilities are

$$p = 1 - p = 0.5$$

Entropy,

$$\begin{aligned} H(p) &= -p\log_2 p - (1-p)\log_2(1-p) \\ &= -0.5\log_2 0.5 - 0.5\log_2 0.5 \\ &= 1 \end{aligned}$$

Hence, the entropy is maximum when $p = 0.5$. When both terms are equally probable, the uncertainty is highest; hence the entropy is maximum. This represents the most information per message.

B&W TV Picture Information Rate 1.19

A B&W TV picture consists of 525 lines of picture information. Assume that each element can have 256 brightness levels. If pictures are repeated at a rate of 32 frame/sec, calculate the average rate of information conveyed by a TV set to a viewer.

Given

- Number of lines per frame = 525
- Brightness levels per picture element = 256
- Frame rate = 32 fps
- Pixels per line = 256 (as used in the note)

Pixels per Frame

$$525 \times 256 = 13440 \text{ pixels/frame}$$

Bits per Pixel

$$\log_2 256 = 8 \text{ bits}$$

Bits per Frame

$$8 \times 13440 = 107520 \text{ bits/frame}$$

Information Rate

$$\begin{aligned} 107520 \times 32 &= 3440640 \text{ bps} \\ &= 3.44 \text{ Mbps} \end{aligned}$$

Finding $H(X)$, $H(Y)$, $H(X \mid Y)$, and $H(Y \mid X)$ 1.20

Calculate $H(X)$, $H(Y)$, $H(X \mid Y)$, and $H(Y \mid X)$ from the data given below.

Y \ X	**1**	**2**	**3**	**4**
1	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$
2	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{32}$	$\frac{1}{32}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
4	$\frac{1}{4}$	0	0	0

From the table above,

$$P(x_1) = \frac{1}{8} + \frac{1}{16} + \frac{1}{16} + \frac{1}{4} = \frac{1}{2}$$
$$P(x_2) = \frac{1}{16} + \frac{1}{8} + \frac{1}{16} + 0 = \frac{1}{4}$$
$$P(x_3) = \frac{1}{32} + \frac{1}{32} + \frac{1}{16} = \frac{1}{8}$$
$$P(x_4) = \frac{1}{32} + \frac{1}{32} + \frac{1}{16} = \frac{1}{8}$$

We have,

$$H(X) = -\sum_{i=1}^{4} P(x_i) \log_2 P(x_i)$$
$$= \frac{1}{2}\log_2 2 + \frac{1}{4}\log_2 2^2 + \frac{1}{8}\log_2 2^3 + \frac{1}{8}\log_2 2^3$$
$$= \frac{1}{2} + \frac{1}{2} + \frac{3}{8} + \frac{3}{8}$$
$$= \frac{7}{4} \text{ bits/symbol}$$

Now,

$$P(y_1) = \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{32} = \frac{1}{4}$$
$$P(y_2) = \frac{1}{16} + \frac{1}{8} + \frac{1}{32} + \frac{1}{32} = \frac{1}{4}$$
$$P(y_3) = \frac{4}{16} = \frac{1}{4}$$
$$P(y_4) = \frac{1}{4}$$

Now,

$$H(Y) = -\sum_{i=1}^{4} P(y_i) \log_2 P(y_i)$$
$$= \frac{4}{4}\log_2 4$$
$$= \log_2 2^2$$
$$= 2 \text{ bits/symbol}$$

Now, joint entropy,

$$H(X,Y) = -\sum_{j=1}^{4}\sum_{i=1}^{4} P(x_i, y_j) \log_2 P(x_i, y_j)$$
$$= \frac{2}{8}\log_2 8 + \frac{6}{16}\log_2 16 + \frac{4}{32}\log_2 32 + \frac{1}{4}\log_2 4$$
$$= \frac{2}{8}\log_2 2^3 + \frac{6}{16}\log_2 2^4 + \frac{4}{32}\log_2 2^5 + \frac{1}{4}\log_2 2^2$$
$$= \frac{6}{8}\log_2 2 + \frac{24}{16}\log_2 2 + \frac{20}{32}\log_2 2 + \frac{2}{4}\log_2 2$$
$$= \frac{6}{8} + \frac{24}{16} + \frac{20}{32} + \frac{1}{2}$$
$$= \frac{27}{8} \text{ bits/symbol}$$

Now,

$$H(X \mid Y) = H(X,Y) - H(Y) = \frac{27}{8} - 2 = 1.375 \text{ bits/symbol}$$

and,

$$H(Y \mid X) = H(X,Y) - H(X) = \frac{27}{8} - \frac{7}{4} = 1.625 \text{ bits/symbol}$$